

STC 122 - Practical 1 instructions

Instructions

File and Submission Requirements

- Answer the questions that follow and save all your code in a single R script.
- Name the script **Practical1.R**, where **Practical1** is the filename and **.R** is the file extension.
- Once you have completed your online submission, ensure the script is uploaded to clickUP using the correct file name, **Practical1.R**.
- Ensure all variables are named correctly. **Remember that variable names are case sensitive.**
- Ensure your code does not contain any syntax errors.
- Any code that is commented out will be treated as rough work and will not be marked.

Preparing for the Online Assessment

- The questions in this document are intended to prepare you for the online submission. Make sure you are comfortable with all questions here, as well as the content covered in the lecture notes, before attempting the online assessment.
- You will have **only ONE attempt** at the online submission, and it must be completed in **ONE sitting**. **You cannot save your progress and continue later. If you exit the attempt before submitting all your answers, you will be unable to resume and will forfeit the marks for the practical.**
- It is your responsibility to ensure you have a stable and sufficient internet connection before starting the online submission.

Academic Requirements

- You must include **detailed** explanations of your working as comments in your code, for **every** question in the assignment.
- You must also cite any resources you made use of while completing the assignment. Refer to the document `commenting_and_citing_in_your_code.pdf` on clickUP for guidance.
- You must complete the Declaration of Generative AI Use form after finishing your online submission. **Failure to submit this form will result in a mark of zero for the practical.**

Deadline

- The deadline for this practical is **Friday, 14 August 2026 at 23:59**.

Questions

Answer Questions 1–13 using the **titles.csv** dataset available on ClickUP.

Question 1

- Find the value of the highest IMDB score. Save your response into a variable called Q1a.
- Find the value of the lowest IMDB score. Save your response into a variable called Q1b.
- Find the value of the 1st quartile of the IMDB score. Save your answer as Q1c.
- Find the value of the 2nd quartile of the IMDB score. Save your answer as Q1d.
- Find the value of the 3rd quartile of the IMDB score. Save your answer as Q1e.
- Based on the five-point summary values calculated above as well as the mean, determine the shape of the distribution of IMDB scores. Save your response as a character string (e.g., "A", "B", "C", or "D") into a variable called Q1f.
 - Skewed to the right.
 - Skewed to the left.
 - Symmetrical.
 - Bimodal.
 - Not enough information to determine.

Question 2

Find the name of the title with the highest IMDB score. Save your response into a variable called Q2

Question 3

- How many titles had an IMDB score **greater than** 8. Save your answer in a variable called Q3a
- What **proportion** of titles had an IMDB score of **at least** 8? Save your answer in a variable called Q3b

Question 4

Create a histogram to investigate the distribution of the IMDB ratings. Use breaks = 50. Give your plot an appropriate main title and axis titles.

Investigate the impact of changing the argument breaks. Consider break values of 5, 10, 50 and 100.

Question 5

Create a subset containing only the movies. Save your response into a variable called Q5

Question 6

Calculate the mean runtime of the movies. Save your response into a variable called Q6

Question 7

Calculate the standard deviation for the runtime of the movies. Save your response into a variable called Q7

Question 8

Calculate the **percentage** of movie releases with a runtime within 1.2 standard deviations of the mean runtime of the movies. Your answer should be a value between 0 and 100. Save your response in a variable called Q8.

Question 9

Create a side-by-side boxplot to compare the IMDB ratings for the movies against the IMDB ratings for the TV shows. Make sure you add a legend to your plot. Give your plot an appropriate main title and axis titles.

Question 10

Which boxplot has the largest spread? Assign the correct option to a variable called Q10.

- A. Movies
- B. TV shows

For example, if the correct option is movies, assign "A" to a variable called Q10. If the correct option is tv shows, assign "B" to a variable called Q10.

Question 11

Create a two-way contingency table for the age certification and the type of the release. Save your response into a variable called Q11

Question 12

How many movies have a “G” rating? Save your response in a variable called Q12

Question 13

Create a bar chart of the age certifications. Give your plot an appropriate main title and axis titles.